

Haben Gebrekidan

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EDUCATION

Johns Hopkins University, M.S.E. in Electrical Engineering

Focus: Medical and Surgical Robotics

Baltimore, MD

Aug 2024 – May 2026

Syracuse University, M.S. in Electrical and Electronics Engineering

Focus: Power Electronics, Optimization, and Control Systems; GPA: 3.83/4.00

Syracuse, NY

Aug 2022 – May 2024

Addis Ababa Science & Technology University

B.S. in Electromechanical Engineering; GPA: 3.77/4.00

Addis Ababa, Ethiopia

Sept 2016 – Sept 2021

EXPERIENCE

MUSiiC Lab / LCSR — Johns Hopkins University

Research Assistant, Medical Robotics

Baltimore, MD

Aug 2025 – Present

- Executed one of four core workstreams in a large-scale ARPA-H-funded PSI project involving 100+ collaborators, focused on image-guided intraoperative robotic assistance for breast cancer surgery.
- Architected modular ROS 2 autonomy and real-time control pipelines, including cooperative and autonomous UR5e motion with impedance and admittance control, for repeatable lumpectomy cavity scanning via multi-node coordination through topics, services, and actions.
- Integrated perception with robotic motion by synchronizing image acquisition with robot trajectories to support multi-view coverage planning and surgical cavity documentation.
- Collaborated with Kitware on custom 3D Slicer application development, integrating surgical visualization, robot control, and workflow validation tools.
- Built Gazebo and RViz simulation environments to model and de-risk robotic behaviors before hardware deployment; identified failure modes, debugged root causes, proposed design improvements, and validated solutions through simulation and real hardware testing, improving overall system reliability.
- Conducted phantom and ex-vivo experiments to characterize robotic workflow performance and accuracy; results presented at ARPA-H site-visit demonstrations and contributing to a first-author manuscript in preparation.

DL Manufacturing

Electrical Engineering Technician

Syracuse, NY

Jan 2024 – May 2024

- Assisted in dock manufacturing and testing, including PCB design for power electronics, microcontroller programming, and prototype development and validation prior to deployment.
- Performed electrical maintenance, wiring diagram verification, and compliance checks against standard requirements, working with motors and drives.

Syracuse University

Teaching Assistant, Introduction to Electrical Circuits

Syracuse, NY

Aug 2023 – Jan 2024

- Led weekly recitation sessions for Introduction to Electrical Circuits, reinforcing course material on circuit analysis, theorems, and fundamental electrical concepts.
- Held office hours and provided hands-on lab guidance, grading assignments and exams while offering feedback to reinforce student understanding.

Dynamic Locomotion & Robotics Lab — Syracuse University

Research Assistant

Syracuse, NY

Sept 2022 – Jan 2023

- Implemented ROS-based SLAM, localization, and navigation packages for TurtleBot mobile robots across simulation and hardware environments, enabling autonomous temperature and air quality regulation throughout large buildings for energy-efficient HVAC control.
- Integrated LiDAR and onboard sensing for mapping, obstacle avoidance, and autonomous navigation.

Carrier Corporation

Embedded Control Engineering Intern

Syracuse, NY

May 2023 – Aug 2023

- Developed a Python application to parse HVAC sensor logs into structured Excel reports, handling 120+ event types and 80+ sensor categories for marine refrigeration systems.
- Debugged a critical DCX package parsing error, traced the root cause, and reported findings to the DCX team; validated sensor outputs using Carrier's internal proprietary tooling.

ARCOM Labs

Electrical Engineering Intern

Syracuse, NY

Jan 2023 – May 2023

- Validated RF amplifiers and power electronics using oscilloscopes, multimeters, and bench instruments; debugged PCB-level faults and replaced components to meet performance specifications prior to release.

- Programmed microchips, performed PCB soldering, and executed pass/fail testing to ensure hardware reliability and quality compliance.

New Era Research & Development Center (NERD's Center)

Addis Ababa, Ethiopia

Researcher

Sept 2020 – Aug 2022

- Designed and 3D printed mechanical components in SolidWorks and assembled CNC X-Y plotter systems, handling full system integration from mechanical design to hardware assembly.
- Programmed Arduino microcontrollers in the Arduino IDE to implement embedded motion control, coordinating stepper motors and hardware interfaces for precise plotter operation.

SELECTED PROJECTS AND TECHNICAL ARTICLES

Robotic Breast Surgical Planning Workflow with 3D Slicer

ROS 2, UR5e, 3D Slicer, OpenIGTLink, Python, Surgical Robotics

- Integrated UR5e robot control, surgical endoscope imaging, and 3D Slicer visualization into a unified clinical workflow, enabling cooperative robot positioning and automated field-scan motions for breast tumor localization and high-resolution cavity imaging.
- Built live demo and interface-testing tools to validate real-time data flow between robotic hardware, imaging devices, and planning software.
- **Technical Article:** Advancing Breast Surgical Planning with 3D Slicer. Kitware Blog, March 2026.

Point-of-Care Medical Image Segmentation Benchmark

PyTorch, U-Net, MedT, UNeXt, Kvasir-SEG

- Benchmarked U-Net, MedT, and UNeXt for point-of-care segmentation across accuracy, efficiency, and latency.
- Developed an accuracy–efficiency score that identified UNeXt as the strongest edge-deployment candidate.
- **Paper:** H. Gebrekidan, et al. *Point-of-Care Segmentation: A Pragmatic Benchmark of Lightweight CNN, Transformer, and MLP-Hybrid Models with a Composite Accuracy–Efficiency Metric*. Accepted at ICBET 2026

Ultrasound-Guided Robotic Impedance Control

ROS, MATLAB, Python, Impedance Control, Ultrasound Imaging

- Implemented real-time impedance control for a UR5e robotic manipulator with stiffness modulation driven by ultrasound-derived signals.
- Validated closed-loop system behavior through hardware experiments, analyzing stability and responsiveness in human-interactive settings.

Data-Driven MPC for Renewable-Rich Power Grid Voltage Control

Python, MATLAB, MPC, OSQP, IEEE 9-Bus System

- Designed a data-driven MPC approach to regulate grid voltage under renewable generation variability and operational constraints.
- Validated the controller on the IEEE 9-bus system, demonstrating real-time feasibility for renewable-rich power system operation.
- **Paper:** H. Gebrekidan, et al. *Data-Driven Model Predictive Voltage Control for an IEEE 9-Bus System with High Renewable Penetration*. IEEE conference paper

TECHNICAL SKILLS

Programming: C++, Python, MATLAB

Robotics & Autonomy: ROS 1/ROS 2, MoveIt, Gazebo, RViz; motion planning and trajectory generation; forward/inverse kinematics; SLAM, localization, and navigation; TF/TF2; UR5e/URScript, TurtleBot4; LiDAR and force/torque sensors; robot–camera synchronization; vision-guided data acquisition; finite-state-machine (FSM) based autonomy

Control & Optimization: impedance and admittance control; task-space control; Model Predictive Control (MPC); optimization-based control; quadratic programming (QP); constraint handling; closed-loop feedback systems; system identification

Perception & Imaging: OpenCV, 3D Slicer; medical image segmentation using deep learning; multi-view imaging; geometry-aware perception; perception-driven planning; phantom-based imaging experiments

Machine Learning: PyTorch; CNNs and Transformer-based models; training and evaluation pipelines; model benchmarking (latency, memory, compute); hyperparameter tuning; data augmentation; GPU-based training; Dice and IoU metrics

Systems & Integration: multi-node robotic system integration; ROS2 node architecture; topic/service/action interfaces; simulation-to-hardware deployment (sim-to-real); sensor integration; TCP/IP communication; system debugging and validation

Tools & Platforms: Linux/Ubuntu; Git, GitHub, GitLab; L^AT_EX; JupyterLab, Google Colab; SQLite; ROS bag data handling; experiment logging and reproducibility

Hardware & Prototyping: Arduino, Raspberry Pi; SolidWorks; Altium Designer; CATIA; Proteus; Multisim

HONORS & AWARDS

Rubenstein Fellowship, Johns Hopkins University

LCSR Fellowship, Johns Hopkins University

SWE ARDC Foundation Scholarship

ISA Aerospace and Test Measurement Division Scholarship

ISA Process Measurement & Control Division Award

Syracuse University Merit Scholarship

Best Academic Achievement Trophy, Addis Ababa Science & Technology University